

AI ODYSSEY

THE LLM DESIGN WORKSHOP — MINI HACKATHON

Day 4 · RAG Mini Hackathon

OFFICIAL RULEBOOK

1. EVENT OVERVIEW

AI Odyssey: The LLM Design Workshop is a multi-day AI learning and building program hosted by the Association for the Advancement of Artificial Intelligence (AAAI) Student Chapter. Day 4 of the program is a focused, time-boxed Mini Hackathon centred on Retrieval-Augmented Generation (RAG) systems.

Participants will design, build, and deploy a complete end-to-end RAG application within approximately six hours, working with a large-scale (15 GB+) domain-specific dataset. The event is designed to test practical RAG engineering skills: data handling at scale, retrieval quality, grounding, deployment, and real-time robustness under hidden evaluation.

2. EVENT FORMAT

Duration

Approximately 6 hours, conducted in a single continuous session on Day 4 of AI Odyssey.

Domains

Each team selects one of five real-world domains. Domain selection is first-come-first-served or allotted by the organising committee, depending on team distribution on the day.

Domain	Dataset Scale	Representative Focus Areas
Energy	>15 GB	Consumption patterns, grid data, conservation & efficiency reports
Agriculture	>15 GB	Crop data, soil/weather records, market & yield reports
Finance	>15 GB	Transactions, market data, regulatory & compliance documents
Defense	>15 GB	Logistics, historical records, declassified strategic reports
Healthcare	>15 GB	Clinical records, research literature, treatment guidelines

Dataset

Each team receives access to a large-scale dataset (greater than 15 GB) corresponding to their selected domain. Datasets are provided directly by the organising committee at the start of the event. Teams may not substitute the provided dataset with an alternative dataset of their own choosing.

3. EXPECTED PIPELINE

Every submission is expected to implement a genuine end-to-end RAG pipeline. At minimum, the following stages must be present and demonstrable:

1. Data Ingestion — loading and organising the provided dataset for processing.
2. Preprocessing — cleaning, normalising, and structuring raw data for downstream use.
3. Chunking — splitting documents/records into retrieval-appropriate segments.
4. Embeddings — generating vector representations of each chunk.
5. Vector Indexing — storing embeddings in a vector database/index for efficient similarity search.
6. Retrieval — fetching the most relevant chunks for a given query.
7. LLM Response Generation — using a language model to generate a grounded answer from retrieved context.
8. Deployment — a live, working, publicly accessible chatbot interface.

Submissions that skip a stage (for example, hardcoding responses instead of performing genuine retrieval) will be scored down under Correct RAG Pipeline and are subject to disqualification if misrepresented as functioning retrieval.

4. USE OF AI TOOLS

The use of AI tools — including but not limited to LLM assistants (e.g., ChatGPT, Claude, GitHub Copilot), AI-based code generation, and AI-assisted debugging — is permitted and encouraged throughout the hackathon.

AI Usage Guidelines

- AI tools may be used for code generation, debugging, architecture advice, and documentation.
- Participants remain fully responsible for understanding, testing, and validating any AI-generated code before submission.
- Every team must be able to explain their pipeline, design choices, and code in the live demo without relying on AI to answer on their behalf.
- Using AI to fabricate results, fake a working demo, or misrepresent pipeline functionality is a violation of the Code of Conduct (Section 10) and grounds for disqualification.

5. SUBMISSION REQUIREMENTS

All teams must deploy their working prototype and submit it via the official AAAI portal:

<https://aaaiportal.vercel.app/>

A complete submission must include:

- A live, publicly accessible deployment link to the working chatbot application.
- A public code repository link (GitHub/GitLab) containing the full source code.
- A short write-up (1 page max, or as specified on the portal) describing the architecture, tools/models used, and any known limitations.
- Team name, member details, and selected domain, as per the portal's submission form.

Submission Deadline

Submissions must be finalised and locked on the AAAI portal by the end of Hour 5:30 of the event (see Section 7, Event Schedule). No submissions will be accepted after the deadline. Late submissions, incomplete submissions, or submissions made outside the official portal will not be evaluated.

7. EVENT SCHEDULE (INDICATIVE)

Time Block	Activity
Hour 0 (Kickoff)	Domain allotment, dataset access provided, rulebook briefing, Q&A
Hour 0 – 1	Data exploration, environment setup, architecture planning

Hour 1 – 4	Core build: ingestion, preprocessing, chunking, embeddings, vector indexing, retrieval + LLM integration
Hour 4 – 5	Deployment to the AAAI Portal, UI polishing, internal testing
Hour 5 – 5:30	Final submission window closes — link + repo locked
Hour 5:30 – 6	Live judging: hidden query evaluation and demo presentations

The organising committee reserves the right to adjust timings on the day of the event. Any change will be announced to all teams simultaneously.

8. EVALUATION CRITERIA

Each deployed chatbot will be evaluated out of 100 marks, based on the breakdown below. Faculty/judges will test each submission using a predefined, undisclosed set of 5 hidden queries designed to probe retrieval accuracy, grounding, and robustness to tricky or adversarial questions.

Component	Marks	Focus Area
Correct RAG Pipeline	20	Ingestion → chunking → embeddings → vector index → retrieval → LLM
Retrieval Quality	20	Relevance and precision of retrieved chunks
Hidden Query Performance	20	Live evaluation using 5 undisclosed judge queries
Answer Accuracy / Grounding	15	Factual correctness, no hallucination, source grounding
Deployment & Working UI	10	Live, functional deployment on the AAAI Portal
Performance / Optimization	10	Response latency, chunking/indexing efficiency
Presentation / Demo	5	Clarity of explanation and live walkthrough
TOTAL	100	

Notes on Evaluation

- Hidden queries are not shared with teams in advance and are identical across all teams within a domain, ensuring fair comparison.
- Answers must be grounded in the provided dataset. Fabricated or hallucinated answers will be penalised under Answer Accuracy/Grounding, even if fluent and confident-sounding.
- Latency is measured as time-to-first-response during live testing; excessive latency will be penalised under Performance/Optimization.
- Judges' scoring decisions are final.

9. BONUS CONSIDERATIONS

Additional marks/recognition may be awarded, at the judges' discretion, for:

- Clean, well-documented, and reusable codebase.
- Thoughtful handling of edge cases (e.g., out-of-scope queries, ambiguous questions, empty retrieval results).
- Creative or unusually effective retrieval strategies (e.g., hybrid search, re-ranking, query rewriting).

- Strong UI/UX in the deployed application beyond a bare-minimum chat interface.
- Successful deployment achieved well within the time limit, allowing for demonstrated iteration/optimization.

6. TEAM & ELIGIBILITY RULES

- Team size: as specified by the organising committee at registration (confirm on the AAAI portal registration page).
- Each team may select only one domain and will work with only the dataset provided for that domain.
- All code, pipeline components, and the deployed application must be built during the official hackathon window. Pre-built solutions from before the event are not permitted.
- Teams may use open-source libraries, pretrained embedding models, and third-party LLM APIs. All external dependencies must be disclosed in the submission write-up.
- Each team member is expected to be able to answer questions about the project during judging.

10. CODE OF CONDUCT & DISQUALIFICATION

To maintain a fair and professional environment, the following are strictly prohibited and will result in immediate disqualification:

- Submitting a non-functional or fake deployment (e.g., static/hardcoded responses presented as live retrieval).
- Plagiarising another team's code or a pre-existing public project without disclosure.
- Attempting to access, view, or influence the hidden evaluation queries in advance.
- Any form of harassment, disrespect toward organisers/judges/other participants, or disruptive behaviour.
- Missing the submission deadline specified in Section 5.

The organising committee's decision on any rule violation or disqualification is final and binding.

11. SUPPORT & CONTACT

For technical issues, dataset access problems, or submission portal queries during the event, participants should reach out to the on-ground AAAI volunteers or the designated event coordinator, as announced at the start of the session.